LLM-Driven Ontology Learning: From Term Extraction to Upper-Level Categorization for building Information System models

Keywords: Ontology, Ontology Learning, Large Language Models, Natural Language Definitions

Abstract: Knowledge-intensive Information Systems support high-value corporate tasks. However, building knowledge models requires manual selection and semantic clarification of terminology — a costly bottleneck in enterprise ontology engineering. This work explores automating conceptual modeling—terminology selection and classification into upper-level meta-types using general-purpose Large Language Models (LLMs) to produce precise semantic artifacts without specialized fine-tuning. We introduce an empirical framework comparing four strategies: LLM + Corpus, LLM Generated, NER Model + Corpus, and TF-IDF, applied to 40 Brazilian Pre-Salt scientific articles. Classification is standardized by leveraging the LLM to generate Natural Language Definitions (NLDs) before categorization into a multi-layered hierarchy integrating Basic Formal Ontology (BFO), GeoCore, and GeoReservoir. Resulting artifacts underwent blind validation by domain experts. Results demonstrate that non-specialized LLMs produce higher semantic quality and precision in term extraction and taxonomic categorization than traditional statistical methods by leveraging NLDs. The LLM + Corpus pipeline proved the most robust strategy. Ultimately, this work presents a validated paradigm for modernizing ontology building through strategic, enterprise-scale automation during conceptual modeling.

1 INTRODUCTION

The design of knowledge-intensive Information Systems depends critically on the quality of their underlying conceptual models. Ontologies, as formal and explicit specifications of a shared conceptualization (Studer et al., 1998), provide the structural backbone for these systems and support various applications in knowledge organization within the enterprise environment. The use of domain ontologies has demonstrated value in enhancing semantic clarity, thereby eliminating ambiguities across data models, software applications, and information retrieval systems. This perceived value increases the number of ontology engineering projects in corporate environments, which are built upon specialized terminology formalized in ontology artifacts.

In this work, we focus on the construction of a domain ontology designed to serve as the core structural backbone for an industrial software application that retrieves and analyze analogous deposits in petroleum pre-salt exploration in the industry — highly productive deep-water reservoirs that represent some of the most significant energy discoveries of the last decades. Based on our ontological framework, reservoir information from diverse heterogeneous sources is stored in a uniform database for similarity analysis.

Despite recent advances, Ontology Engineering still faces challenges in optimizing the ontology-building process. Figure 1, extracted from the widely used NeOn Methodology for ontology engineering (Suárez-Figueroa et al., 2011), illustrates the entire ontology construction scenario. The steps of terminology elicitation and clarification of the particular semantics of the vocabulary remain bottlenecks in the projects, since they require the involvement of highly specialized human resources for extended labor sessions. This challenge is even more critical for enterprise applications. We aim to automate or reduce human intervention in the steps "Non Ontological Resource Reuse", "Non Ontological Resource Reengineering", and "Conceptualization" as defined in the NeOn methodology.

Non-ontological resource reuse begins with selecting terminology sources that determine the entities to be modeled. To ensure semantic interoperability and future expansion, these vocabularies are defined within a hierarchical structure (Guarino, 1998): spanning from *top-level ontologies* (foundational categories) through *core ontologies*, down to specialized *domain ontologies*. Therefore, we aim to build a robust domain ontology specialized from both a top and a core ontology.

However, populating this robust framework traditionally relies on manual knowledge acquisition — a laborious bottleneck that cannot keep pace with the exponential growth of unstructured data. While Large Language Models (LLMs) offer a potential so-

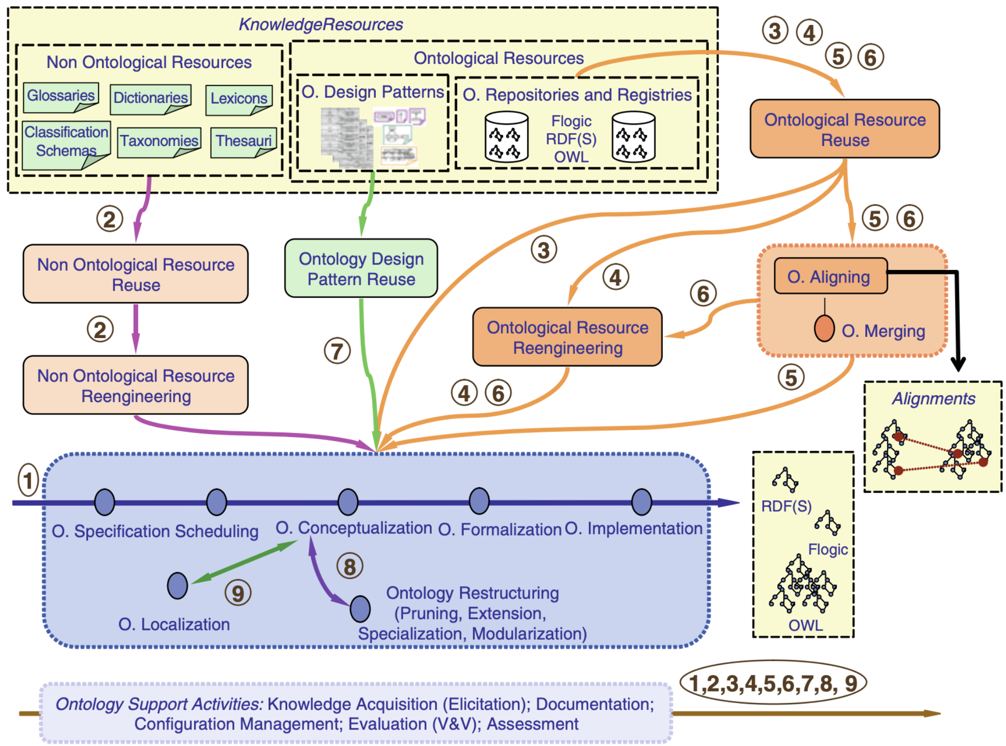

Figure 1: NeOn scenarios for building ontologies. Adapted from (Suárez-Figueroa et al., 2011), p. 13

lution to accelerate this process, approaches relying on domain-specific fine-tuning (Sousa et al., 2023), (Lo et al., 2024) often shift the resource burden, necessitating the curation of costly, manually annotated datasets and demanding substantial computational infrastructure.

This work investigates a more accessible and generalizable path: applying a general-purpose foundation model without domain-specific fine-tuning. We raise a critical research question: Can a non-specialist LLM produce semantically precise artifacts, such as term extraction and categorization into formal, multi-layered ontological hierarchies, effectively reducing human intervention? We design and compare four distinct pipelines for term extraction (1) Corpus-Extractive LLM, (2) LLM Generated, (3) NER Model + Corpus, and (4) a traditional TF-IDF baseline. We then pivot on the LLM's capacity to generate high-quality Natural Language Definitions (NLDs) to improve the automated classification process (Lopes Junior, 2025) by using the LLM itself. The framework is applied to the complex domain of Brazilian Pre-Salt petroleum geology using a curated corpus. The results are then subjected to a blind empirical validation by domain experts to assess quality, leading to a preliminary conceptual artifact based on terms deemed relevant and correctly categorized.

The remainder of this work is organized as follows: Section 2 reviews foundational literature; Section 3 details our comparative empirical approach; Section 4 describes the four pipelines, NLD generation, categorization into upper-level ontologies, and blind validation steps; Section 5 presents the empirical findings and discussion; and Section 6 concludes the study, summarizing insights and next steps.

2 BACKGROUND

This section establishes the foundational knowledge necessary for understanding the proposed approach and reviews the current state-of-the-art literature relevant to LLM application in ontology engineering.

2.1 Ontology Learning

Ontology Engineering's primary bottleneck remains knowledge acquisition — the manual, expert-driven process of identifying relevant concepts, relations, and axioms from domain-specific sources. This has led to the sub-field of Ontology Learning, which seeks to automate this process, primarily from unstructured text (Khadir et al., 2021). This process can be modular, using specialized tools for individual components. Our work focuses on the foundational upstream tasks of term extraction and categorization, reviewing the evolution of methods for these specific tasks.

2.2 Ontology Learning and Machine Learning

Traditional automated approaches to term extraction have relied on statistical methods, such as the one proposed by (Garcia et al., 2020b), which utilized frequency-based analyzes (e.g., TF-IDF) to extract and rank terms from a domain corpus. This work underscored the continued, critical role of domain expert validation. However, these methods fail semantically, unable to separate critical domain concepts from noise

Embedding-based methods progressed from early WordNet-based (Miller, 1994) approaches (Schmidt et al., 2019), (Schmidt, 2020) that were limited by general-purpose embeddings. The subsequent introduction of contextual embeddings (e.g., BERT (Devlin et al., 2019)) enhanced semantic capability for downstream tasks like alignment and subsumption prediction (Chen et al., 2023), (He et al., 2022), including methods utilizing top-level ontology context (e.g.,, the rdfs:comment field) (Sousa et al., 2023) to improve automatic classification of domain concepts. However, these specialized models still require costly domain-specific fine-tuning and manually labeled data, reintroducing a significant human bottleneck.

2.3 Ontology Learning and Large Language Models

Systematic reviews confirm the significant impact of deep learning and LLMs across the ontology engineering lifecycle, from requirements specification to maintenance (Amalki et al., 2025). This growing trend underscores the potential of LLMs to reduce the heavy reliance on human domain experts (Kommineni et al., 2024).

Different LLM model families were evaluated for term typing, taxonomy discovery, and non-taxonomic relation extraction in zero-shot settings by (Babaei Giglou et al., 2023), indicating that in their pre-trained state, LLMs performed worst on term typing overall and struggled with ontology construction. The LLMs4OL 2024 Challenge (Giglou et al., 2024) established the first community-wide benchmarking initiative for LLMs in ontology learning, further validating the field's growing commitment to systematic evaluation.

Fine-tuning approaches have emerged as a relevant strategy. (Lo et al., 2024) introduced an end-to-end ontology learning method (*OLLM*) that fine-tunes LLMs with custom regularization to reduce overfitting on high-frequency concepts, avoiding the limitations of subtask-based composition. However, fine-tuning requires substantial labeled training data and domain-specific annotation effort, which may not be feasible in all ontology engineering contexts.

The capacity of LLMs to generate OWL ontology drafts directly from ontological requirements was investigated by (Saeedizade and Blomqvist, 2024), highlighting that advanced LLMs can produce OWL suggestions comparable to those of novice human modelers.. Aligning with these findings, (Lippolis et al., 2025b) evaluated the generation of OWL ontology drafts from ontological requirements through the usage of advanced prompting techniques, significantly outperforming novice human engineers in ontology modeling results, but still indicating the necessity for human oversight. *NeOn-GPT* was introduced by (Fathallah et al., 2024): a semi-automatic ontology learning pipeline that combines the structured *NeOn* ((Suárez-Figueroa et al., 2015)) methodology with the generative capabilities of LLMs. The authors highlight key limitations in its ability to produce formally expressive ontologies that are validated against the gold standard ontology used in the study. Three prompting strategies - direct, sequential, and sentence-by-sentence - were explored by (Bakker et al., 2024), observing that while LLMs effectively identify classes and individuals, they often struggle to generate consistent properties.

High performance (over 90% macro F1-score) and robustness of language models in classifying domain entities into top-level ontology concepts using Natural Language Definitions (NLDs), even across multilingual settings, was demonstrated by (Lopes Junior, 2025). This work highlights that NLDs are the most effective textual representation for this classification task. This critical insight provides a path forward: no work has yet been designed or empirically compared to leverage this NLD-centric insight for the engineering task of extracting domain terms from a corpus and classifying them against a formal, multi-layered upper-level ontology hierarchy.

3 A NEW APPROACH FOR TERM EXTRACTION AND CLASSIFICATION

The path forward follows the insight from (Lopes Junior, 2025) that classification is highly effective if provided with a high-quality Natural Language Definition. We designed an empirical experiment comparing four distinct term extraction pipelines: (1) LLM + Corpus (2) LLM Generated, (3) NER + Corpus, and (4) TF-IDF as a baseline.

All four pipelines converge on a single, standardized classification module that (1) leverages an LLM to generate a Natural Language Definition (NLD) for each term, and then (2) uses that NLD to perform a complex categorization against a formal, multi-layered upper-level ontology structure. This structure is built upon three integrated ontologies: BFO (Basic Formal Ontology) (Arp et al., 2015), which serves as the foundational ontology; GeoCore (Garcia et al., 2020a), a core ontology in the domain that formally utilizes BFO; and GeoReservoir (Cicconeto et al., 2022), a parallel domain ontology built upon GeoCore. GeoReservoir has a highly significant overlap with the domain for which we intend to create this new ontology, meaning it will likely serve a complementary role. The BFO and GeoCore were a natural choice to expand the approach of (Lopes Junior, 2025) and ensure further integration with the existing geology ontologies.

Then we execute a blind, multi-expert empirical validation. This assessment quantitatively and qualitatively evaluates the final conceptual artifacts (Terms, NLDs, and Categories) from all pipelines.

4 METHODOLOGY

We have organized the methodology of this work into four phases. The first phase is the preparation of the curated corpus. The second phase consists of term acquisition (performed using four distinct pipelines). The third phase involves generating natural language definitions (NLDs) and classifying the terms. Lastly, the fourth phase consists of a comparative assessment of the four pipelines across three critical dimensions: term extraction coverage, NLD accuracy, and taxonomic categorization.

4.1 Phase 1: Corpus Preparation

The foundation of our experiment is a highly specialized corpus consisting of 40 curated scientific articles in the field of the Brazilian Pre-Salt. A senior domain specialist rigorously selected these articles to ensure that the corpus represents the maximum of optimized and authoritative knowledge available. This focused approach ensures the corpus is peer-reviewed, representative, and authoritative for the subsequent term extraction and classification tasks, providing the necessary informational density while aligning with the limited human resources.

The method automatically extracted the text from all documents using a simple Python script leveraging the open-source library `pdftotext`. We decided to retain the entire extracted text, including abstracts and references, because these sections often contain high-value terms and concepts definitions critical for ontology construction.

4.2 Phase 2: Competing Term Acquisition Pipelines

All LLM-based steps utilized the general-purpose, non-specialized foundation model Gemini 2.5 Pro throughout the process. This choice was justified by both its advanced reasoning capabilities, which are crucial for categorization tasks, and by its high availability, ease of use, and streamlined API access, ensuring robust, reproducible, and operationally efficient execution. Unless otherwise specified (i.e., Pipeline 3), we have maintained a temperature of 0 for all LLM steps to maximize output determinism.

- **Pipeline 1 (Corpus-Extractive LLM):** This pipeline uses the LLM in a purely extractive mode. We prompted the model to read the text of each of the 40 articles and extract all relevant geological terms. The method aggregated the extracted terms through stemming before proceeding to the subsequent classification workflow.
- **Pipeline 2 (LLM Generated):** This pipeline operates in a zero-shot, corpus-agnostic manner. We prompted the LLM to generate a list of the 500 most important terms for the domain (Pre-Salt and analogs) based solely on its internal, pre-trained knowledge. A low, non-zero temperature (T=0.2) was chosen as a principled heuristic. This option introduces minimal stochasticity, allowing the model to explore a slightly broader range of high-probability terms and mitigate the risk of repetitive, overly deterministic output, while maintaining a strong focus on factual relevance.
- **Pipeline 3 (Hybrid NER-LLM):** This pipeline tests a hybrid, cross-lingual approach. The specialized multilingual NER model XLM-RoBERTa Large (Moreira et al., 2025)) was al-

lowed to extract terms and then fine-tuned on the PetroGeoNER petroleum-specialized corpus. Then, the method processed the terms and their respective NER labels, which were extracted by this model.

- **Pipeline 4 (Statistical Baseline):** This pipeline replicates the traditional methodology (Garcia et al., 2020b) and serves as our non-AI baseline. It uses the TF-IDF (Term Frequency-Inverse Document Frequency) weighting strategy, one of the most popular and effective methods in information retrieval. Our method ran on the 40-article corpus.

The raw outputs from all four pipelines showed significant variability. To ensure systematic and reproducible aggregation, a normalization and consolidation step was implemented. This process involved applying a stemmer for English to normalize terms to their root form, followed by mapping each stem back to its most frequent lexical form for human readability. The steps produced four distinct, consolidated, and frequency-filtered lists, establishing the robust empirical basis for the subsequent NLD-centric classification workflow.

The results from all distinct pipelines, along with the source code and the specific LLM prompts used for each step, are made available on https://anonymous.4open.science/r/PreSaltOntology-61FA/

4.3 Phase 3: NLD-Centric Classification Workflow

This phase implements a single, standardized workflow to process the candidate terms from all four pipelines. We designed this workflow to explicitly test the findings of (Lopes Junior, 2025) by using NLDs as the pivot for classification.

All unique candidate terms from the four consolidated lists were fed into an NLD generation module, leveraging the LLM to generate a canonical definition. We prompt the LLM with a system instruction defining its role as a senior geoscientist and ontology engineer. For each term, the LLM receives a prompt template instructing it to generate a concise, technically clear, Aristotelian-style definition (e.g., "X is a Y that Z") aligning with (Lopes Junior, 2025) findings.

The method converted the generated pairs (Term, NLD) into JSON arrays and sent them to a dedicated LLM classifier. Receiving one request per term, a Chain-of-Thought (CoT) prompt instructed the model to analyze the term and its NLD, and then classify it into the most specific category possible. This classification strictly followed a fallback hierarchy: first, GeoReservoir (more specialized ontology), then GeoCore (core ontology), and finally BFO (the top-level ontology).

4.4 Phase 4: Specialist Evaluation

To evaluate the performance of each pipeline, we designed a blind, multi-expert empirical validation performed by geologists specialized in the geological Pre-Salt domain.

- **Expected terms:** Before any sequential steps, five domain specialists provided a reference set of 100 terms (20 terms each) representing essential concepts for the Pre-Salt domain. This group consisted of highly experienced professionals with advanced academic degrees (PhD and Master's) and extensive combined experience spanning both petroleum industry operations and petroleum academia.
- **Sampling:** The method created a final sample from each pipeline by selecting the 100 most relevant terms, along with their generated NLDs and assigned categories, from each of the four pipelines, based on its respective relevance ranking mechanism. These 400 entries were combined, anonymized, and randomized into a single list. This fixed limit of 100 terms per pipeline was a pragmatic decision driven by limited human resources and the recognized labor-intensive nature of expert validation, ensuring the empirical assessment remains feasible, rigorous, and non-exhaustive for the domain specialists.
- **Quantitative Metrics:** The key metrics collected for this study were:
  - **Term Relevance:** To measure the quality of the term acquisition (Relevant, Irrelevant, Invalid, Unknown). The same five domain specialists that provided the Expected terms participated in this evaluation.
  - **Definition Accuracy:** To measure the quality of the NLD-generation step (Correct, Partially Correct, Incorrect, Ambiguous / Too Vague, Wrong Context, Unknown). Three from the previous five domain experts participated in this validation step: one highly specialized researcher, with over 20 years of experience in the Brazilian Pre-Salt domain; one PhD specialist in Pre-Salt reservoir characterization with 17 years of industry experience; and one Master's degree holder, also active in the Pre-Salt field.

Table 1: Term Extraction Coverage Analysis (Recall %))

Pipeline	Recall (%)
LLM + Corpus	60
LLM Generated	40
NER + Corpus	32
TF-IDF	25

Table 2: Extracted Terms Relevance Distribution per Pipeline

Pipeline	Relevant (%)	Irrelevant (%)	Unknown (%)	Invalid (%)
LLM + Corpus	97,8%	2,0%	0,2%	0,0%
LLM Generated	96,2%	3,8%	0,0%	0,0%
NER + Corpus	86,6%	11,8%	0,8%	0,8%
TF-IDF	46,0%	46,4%	1,4%	6,2%

- **Category Accuracy:** To measure the quality of the final classification, using a metric to capture the accuracy and specificity (e.g., Correct and Specific, Correct but Unspecific, Incorrect, Unknown). The same three specialists from the previous step participated in this evaluation.

This mixed-methods assessment enables an empirical comparison of the results from each pipeline.

5 RESULTS AND DISCUSSION

This section presents the comparative evaluation of the four ontology learning pipelines: LLM + Corpus, LLM Generated, NER + Corpus, and TF-IDF. Based on the assessment by domain specialists, we analyze performance across four critical dimensions: term extraction coverage, terminological relevance, NLD accuracy, and taxonomic categorization accuracy.

5.1 Term Extraction Coverage

The terms extracted by the pipelines were compared against the expert-defined reference terms (20 terms provided by each of the five specialists) deemed essential for the domain coverage. The LLM + Corpus pipeline achieved the highest coverage, followed by LLM-generated, while NER + Corpus and TF-IDF demonstrated significantly lower recall (Table 1).

A clear distinction emerged in entity type performance. While concrete geological objects (e.g. 'Calcite', 'Coquina') were widely detected by LLM and NER pipelines, a systematic gap existed in the extraction of petrophysical properties and attributes (e.g. 'grain roundness'). This absence was attributed to strict prompt constraints prioritizing "conceptual classes" and excluding numerical values, confirming model adherence rather than capability limitations. Future iterations should explicitly request properties to mitigate this.

A "granularity gap" was also noted, where automated extraction favored generic terms over the specific compound terms that experts had expected. Notably, TF-IDF exhibited the lowest efficacy, failing to capture low-frequency terms that are highly relevant (e.g., 'Stevensite'). In contrast, semantic LLM methods successfully retrieved these terms.

In summary, the LLM + Corpus pipeline proved the most robust for conceptual entities, offering the best balance between precision and abstraction. While NER + Corpus showed resilience in detecting named entities filtered out by LLM prompt constraints, it lacked semantic depth.

5.2 Term Relevance Evaluation

Table 2 reveals a distinct dichotomy between semantic (LLM-based) and statistical/syntactic (TF-IDF/NER + Corpus) extraction methods. The LLM + Corpus pipeline achieved the highest overall and consensus relevance. This result contrasts sharply with the TF-IDF baseline, which had introduced significant noise (46.4% Irrelevant, 6.2% Invalid).

Furthermore, the analysis of evaluator consensus in Table 3 highlights a qualitative advantage of LLMs. While the NER + Corpus pipeline achieved a respectable relevance rate, it generated a high volume of 'Mixed Feelings' - this category signifies terms where evaluators lacked consensus, being deemed simultaneously relevant and irrelevant by different specialists.

Finally, a subtle but important distinction emerged between the LLM approaches. While both methods achieved high relevance (¿96%), the LLM + Corpus pipeline proved superior by using source text as a se-

Table 3: Term Relevance Consensus Metrics

Pipeline	Strictly Relevant	Strictly Irrelevant	Mixed Feelings
LLM + Corpus	91	0	9
LLM Generated	88	0	12
NER + Corpus	66	1	33
TF-IDF	25	22	53

Table 4: Natural Language Definition Evaluation Overall

Accuracy Status	LLM + Corpus	LLM Generated	NER + Corpus	TF-IDF
Correct	53,7%	66,0%	57,0%	34,7%
Partially Correct	34,0%	24,3%	22,3%	22,0%
Incorrect	9,7%	6,7%	15,3%	9,0%
Ambiguous or Too Vague	1,0%	2,3%	2,0%	14,7%
Unknown	1,0%	0,3%	2,0%	15,7%
Wrong Context	0,7%	0,3%	1,3%	4,0%

mantic anchor, resulting in slightly but significantly higher consensus (91 for LLM + Corpus against 88 for LLM-generated). It contributed to mitigating the ”domain blindness” of the LLM-generated approach, demonstrating that corpus grounding can help capture the specialized vocabulary of the Pre-Salt domain.

5.3 Natural Language Definition Evaluation

Considering the identical, context-devoid definition prompt across all pipelines, performance divergence stemmed solely from the input term nature. The LLM Generated pipeline achieved the highest accuracy 4. This was attributted to the model ”internal coherence”, where the model prioritized canonical, theoretical concepts (e.g., ’Dolomite’) for which it possesses robust internal definitions.

The LLM + Corpus pipeline had a high rate of ’Partially Correct’ definitions. This occurred because complex, empirically extracted terms lacked the source context necessary for the general LLM to generate precise definitions, leading to generic outputs insufficiently nuanced for experts. This exposes the critical methodological limitation of context blindness during the definition phase. These findings demonstrate that defining empirically extracted terms is significantly more demanding than defining theoretical ones. Expert feedback also highlighted a scientific currency limitation: definitions sometimes reflected outdated paradigms (e.g., microbial origin of Pre-Salt carbonates), due to the LLM’s training data’s temporal constrains. Future iterations must incorporate Retrieval-Augmented Generation (RAG) to address both issues by providing necessary context and an updated stream of high-quality domain literature to prevent obsolete paradigm propagation.

5.4 Taxonomic Categorization Evaluation

Taxonomic categorization evaluation results are shown in Table 5. It reveals a fundamental trade-off between terminological stability and domain validity driven by the origin of the input terms. The LLM Generated pipeline achieved the highest Strict Consensus. The model prioritized canonical, theoretical concepts (e.g., ’Dolomite’) from its pre-training that elicit unanimous expert agreement. Conversely, the LLM + Corpus pipeline demonstrated superior overall accuracy, achieving the highest Majority Consensus with the lowest rejection rate. While real-world terms extracted from reports often exhibit complexity that sparks taxonomic debate, they represent a more accurate reflection of the Pre-Salt domain than the theoretically ”clean” of the generative approach, which saw a higher rejection rate.

The significant proportion of terms achieving Not Strict Consensus over Strict Consensus (Correct and Specific) in LLM + Corpus pipeline (Table 6) exposes a critical methodological bottleneck: context blindness. This suggests the LLM’s full reasoning potential is currently constrained by the data architecture. This may also indicate that implementing RAG in the NLD generation phase can improve categorization, and elevate majority consensus instances to strict consensus. Additionally, the limitations of non-semantic methods were evident. NER + Corpus showed high disagreement (45% Majority Correct vs. 23% Strict Correct and Specific), and TF-IDF yielded the high-

Table 5: Category Accuracy Distribution

Accuracy Status	LLM + Corpus	LLM Generated	NER + Corpus	TF-IDF
Correct & Specific	67,00%	64,67%	61,67%	40,67%
Correct but Inespecific	15,00%	11,00%	20,67%	10,00%
Incorrect	16,33%	22,67%	14,33%	25,33%
Unknown / Cannot Judge	1,67%	1,67%	3,33%	24,00%

Table 6: Category Accuracy Consensus Metrics

Consensus Metric	LLM + Corpus	LLM Generated	NER + Corpus	TF-IDF
Strict Consensus:				
Correct and Specific	39	43	23	17
Incorrect	3	9	3	4
Correct but Inespecific	1	1	0	0
Unknown	0	0	0	0
Not Strict Consensus:				
Correct and Specific	32	23	45	19
Incorrect	8	14	6	17
Correct but Inespecific	10	8	10	5
Unknown	0	0	0	16
Ambiguity:				
No consensus	7	2	13	22

est rate of ”Unknown” results.

The high acceptance in the Table 7 rate further confirms that using the corpus acts as a decisive ’Validity Filter’ for the extracted terms. When accounting for partial success, the LLM + Corpus pipeline achieved a total acceptable categorization rate (Majority Correct and Majority Partially Correct) of 71 and 11, underscoring its efficacy as the primary knowledge acquisition tool when compared to LLM-generated categorized terms.

5.5 Resulting Conceptual Artifact

The experiment also produced a tangible output: a set of validated conceptual artifacts that represent a preliminary domain ontology for Brazilian Pre-Salt petroleum reservoirs, as shown in Figures 2 and 3. This preliminary pre-salt ontology is available at https://anonymous.4open.science/r/PreSaltOntology-61FA/

The final consolidation of results yielded a core set of domain concepts that achieved unanimous consensus among the three expert evaluators regarding both Term Relevance (Relevant) and Category Accuracy (Correct and Specific). These terms were formally instantiated into our ontological hierarchy.

Although the generated ontology is not axiomatized and does not encompass the entire set of required dependent entities, such as properties or relationships, this resulting ontology remains significant. It demonstrates that a semi-automated pipeline, despite its current limitations, when properly designed with NLD generation and expert-in-the-loop validation, can bridge the gap between raw text and formal ontological structures as a short step in ontology engineering. It moves beyond term extraction to produce semantically grounded, hierarchically consistent artifacts, thereby easing the most labor-intensive initial phases of domain ontology construction.

6 CONCLUSION

This study evaluated a methodological framework for partially automating Ontology Engineering knowledge acquisition in specific domains. Comparing four pipelines, the LLM + Corpus approach proved most robust for term extraction, outperforming TF-IDF and NER significantly. Furthermore, the novel application of NLDs as a pivot for automated classification proved effective, enabling the mapping of raw terms into a formal, multi-layered hierarchy (BFO, GeoCore, GeoReservoir).

Beyond the methodological insights, this process yielded a tangible, expert-validated conceptual artifact that addresses the specific needs of the Brazilian Pre-Salt domain. This ontology serves as a structural foundation for a enterprise software application in the

Table 7: Final Category Accuracy: Majority Consensus (%)

Majority Metric	LLM + Corpus	LLM Generated	NER + Corpus	TF-IDF
Majority Correct	71	66	68	36
Majority Partially Correct	11	9	10	5
Majority Disputed / Rejected	18	25	22	59

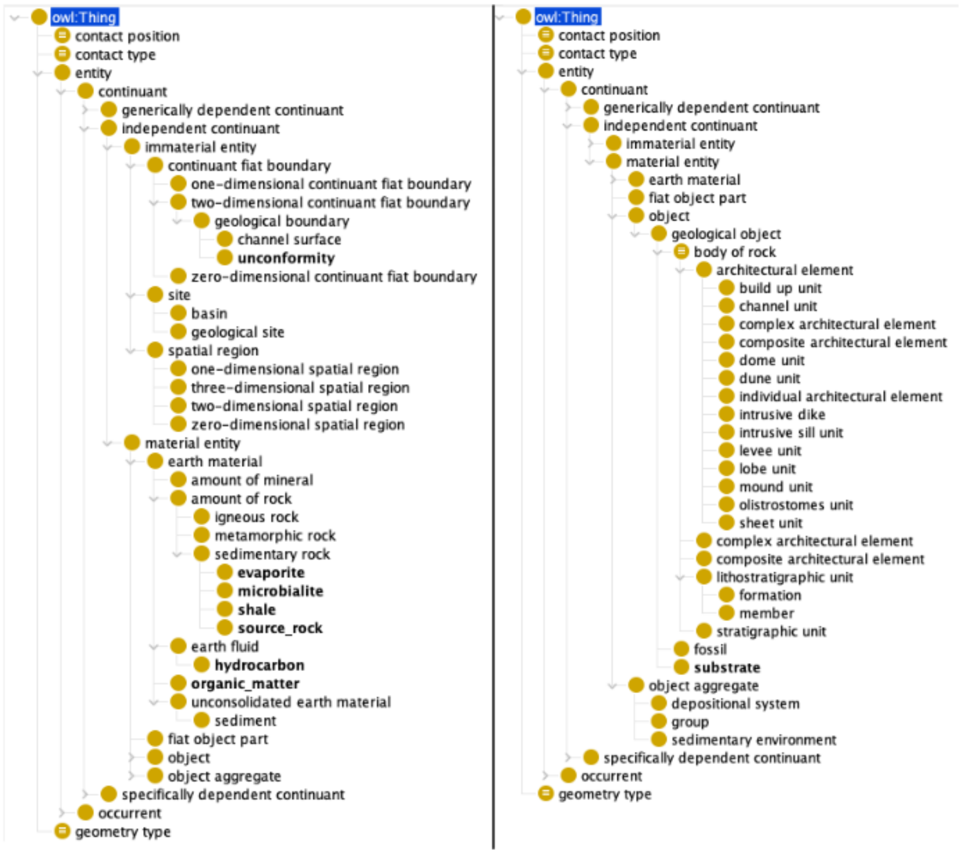

Figure 2: Resulting conceptual artifact, containing the validated terms classified into the three hierarchical levels: BFO, GeoCore, and GeoReservoir. Classes in bold indicate the extracted terms. In the left tree, Unconformity is subsumed by GeoCore:GeologicalBoundary; Evaporite, Microbialite, Shale and Source Rock are subsumed by GeoReservoir:SedimentaryRock; Hydrocarbon is subsumed by GeoCore:EarthFluid; and Organic Matter is subsumed by GeoCore:AmountOfRock. In the right tree, Substrate is subsumed by GeoReservoir:BodyOfRock.

petroleum industry, designed to retrieve and analyze analogous petroleum deposits, enabling the integration of heterogeneous reservoir data into a unified, queryable format.

Empirical validation revealed a trade-off: while LLM Generated pipeline provided better definitions ("internal coherence") but less relevant terms, LLM + Corpus provided more relevant results but struggled with ambiguous terms and outdated paradigms. Future work will integrate RAG into the NLD generation phase to overcome this context blindness. RAG is expected to improve reasoning to resolve ambiguity and ensure scientific currency, paving the way for a scalable, semi-automated process.

While this article presents an overview of the results, we have also made available all supplementary material, including the code, curated corpus, the full list of categorized terms, as well as the corresponding expert evaluations in https://anonymous.4open.science/r/PreSaltOntology-61FA/.

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Figure 3: Resulting conceptual artifact, containing validated terms classified into the three hierarchical levels: BFO, GeoCore, and GeoReservoir. Classes in bold indicate the extracted terms. In the left tree, Alkalinity, Permeability, Reservoir Quality and Porosity are subsumed by BFO:Quality; Onlap is subsumed by GeoCore:GeologicalContact. In the right tree, Compaction, Diagenesis, Dissolution, Erosion, Evaporation, Silicification and Subsidence are subsumed by GeoCore:GeologicalProcess; Hauterivian, Early Cretaceous, Cretaceous, Barremian, Aptian and Albian are subsumed by GeoCore:GeologicalTimeInterval.

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